Doyun (Logan) Kim

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Education

University of Washington Seattle, WA
Bachelor of Science in Electrical and Computer Engineering – GPA: 3.59 /4.0 *Exp. Grad. Dec 2026*

Skills

Programming: Python, C, C++, Java, MATLAB, SystemVerilog, Bash

Tools: Fusion 360, KiCAD , VSCode, Arduino IDE, git, ROS, Docker, Linux (Ubuntu), LTspice, ModelSim, MATLAB Simulink, Gazebo, Microsoft Office Suite

Technical Skills: 3D printing, CAD Modeling, Schematics, PCB layout, Soldering

Experience

Cyberworks Robotics Inc. Dec 2025 – Jun 2026
Electrical Engineer *Seattle, WA*

- Developed and validated a ROS1-based autonomous navigation pipeline integrating YOLOv8, OpenVINO, RGB-D sensing, TF transforms, and custom binary heuristics for real-time crowd-aware wheelchair navigation on embedded CPU hardware.
- Achieved real-time embedded perception performance of approximately **30–50 FPS** using YOLOv8 Nano inference on NUC hardware while maintaining **sub-100 ms** detection latency targets for closed-loop robotic decision-making.
- Implemented and debugged robotic simulation and control infrastructure using Gazebo PedSim to test autonomous navigation, obstacle avoidance, SLAM, and motion-control behaviors in dense pedestrian environments.
- Revived and integrated a powered Quantum Q6 Edge wheelchair platform by redesigning onboard power distribution, implementing DC/DC buck converters for sensors and computing hardware, and integrating peripheral modules for autonomous mobility experimentation.

UW Solar Vehicle Team Sep 2025 – Jun 2026
Electrical Engineer *Seattle, WA*

- Designed a high-voltage power-on circuit to safely energize the car's electrical system from a 126 V nominal main battery, incorporating a precharge circuit and shunt for current measurement integrated into STM32-based switching logic.
- Designed a 24 V power distribution board in KiCAD that delivers battery power across 8 independent electrical subsystems via a 4-layer PCB with traces sized for their respective load currents.
- Simulated and validated circuit behavior in LTspice prior to fabrication; prototyped boards to verify functional performance and document stress limits under load.

US 8th Army Korean Augmentation to the United States Army June 2022 – October 2023
Signals Intelligence Voice Interceptor(35P) *Pyeongtaek, South Korea*

- Led a team of 7 SIGINT mission specialists composed of 3 Korean Army soldiers and 4 U.S. Army soldiers.
- Established an efficient communication channel between two intelligence commands across the ROK Army and the U.S. Army.
- Acquired security clearance and disseminated Secret-level intelligence by intercepting and transcribing foreign communications
- Operated and maintained the signal processing software that handle communication traffic with different radio frequencies.

Projects

LAIKA, the Quadrupedal Robot

- Designed and 3D-printed a 12-body-part quadrupedal robot modeled in Fusion 360; validated servo motors and sensors using custom Python test scripts.
- Conducted power analysis to achieve the operation time of **~22 minutes** at the typical current draw of 13.25 A.
- Derived DH parameters for forward kinematics and implemented gait control logic via inverse kinematics; simulated locomotion and real-world navigation in MATLAB Simulink.
- Deployed a ROS-based framework on an Ubuntu-booted Raspberry Pi for modular and efficient system organization.

Face Tracking Turret

- Designed a 3 DOF face-tracking turret capable of stable detection and tracking of faces at ~30 FPS controlled with ESP32.
- Configured serial connection between the Python agent and ESP32 with PySerial to achieve real-time update of servos.
- Implemented face detection with OpenCV's Haar Cascade classifier and precise tracking in servos with PD control.
- Optimized joint movement using deadband and rate-limiting to prevent servo overshoots, ensuring smooth turret motion.